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NeuroHack 2026

SPECTRA-ICA: Spectral Profile-Estimated Continuous Temporal Removal of Artifacts via ICA

- Anand Lo
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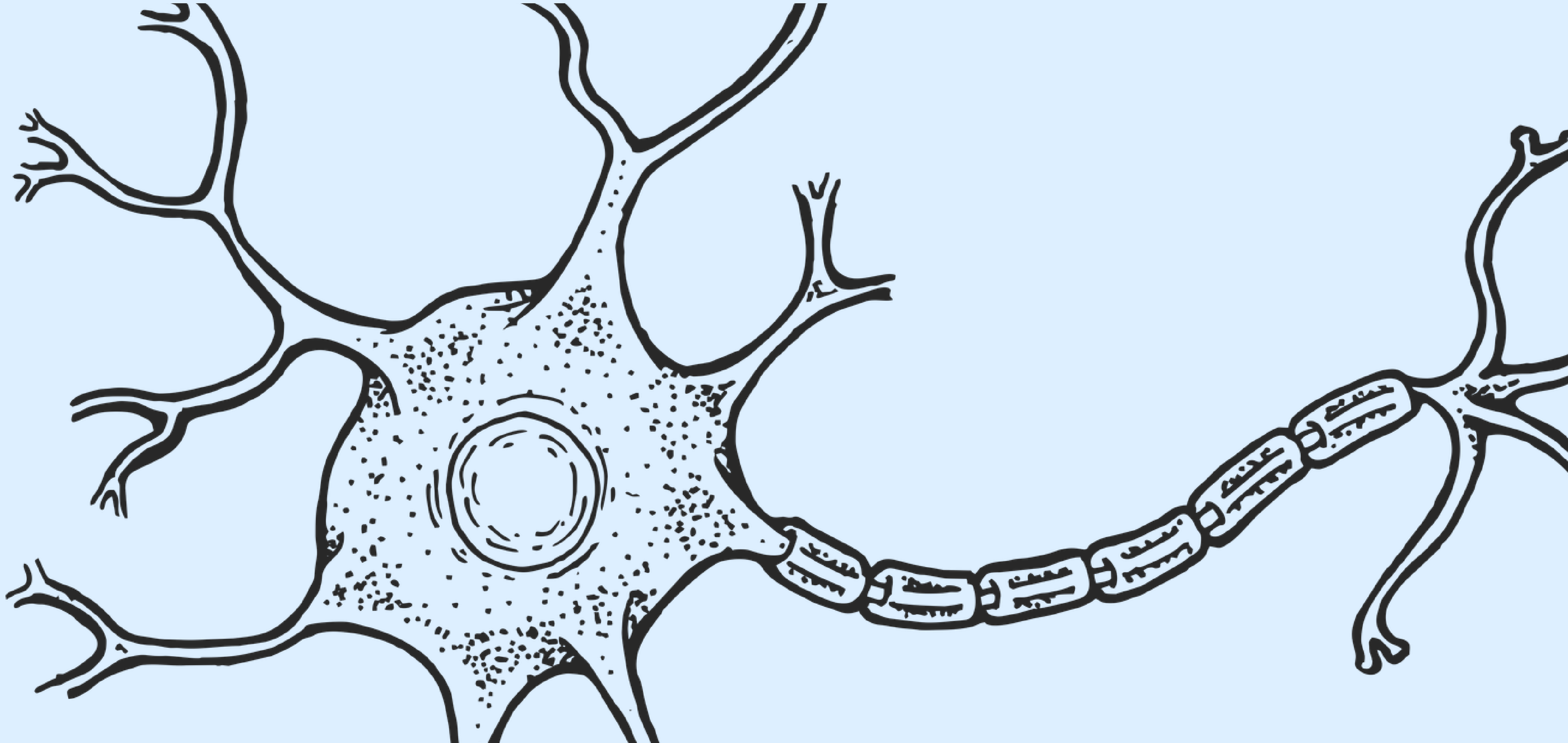
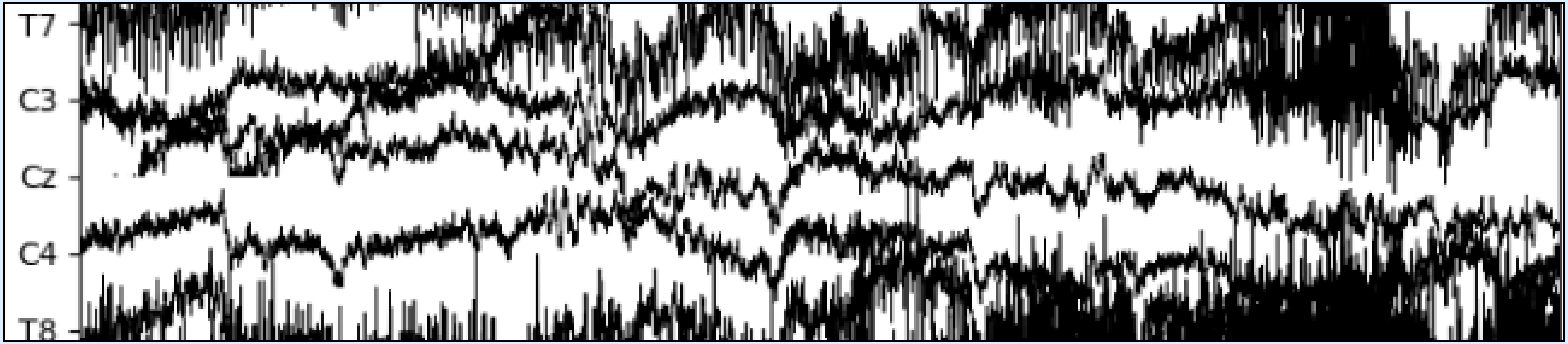
- Milad Alipour



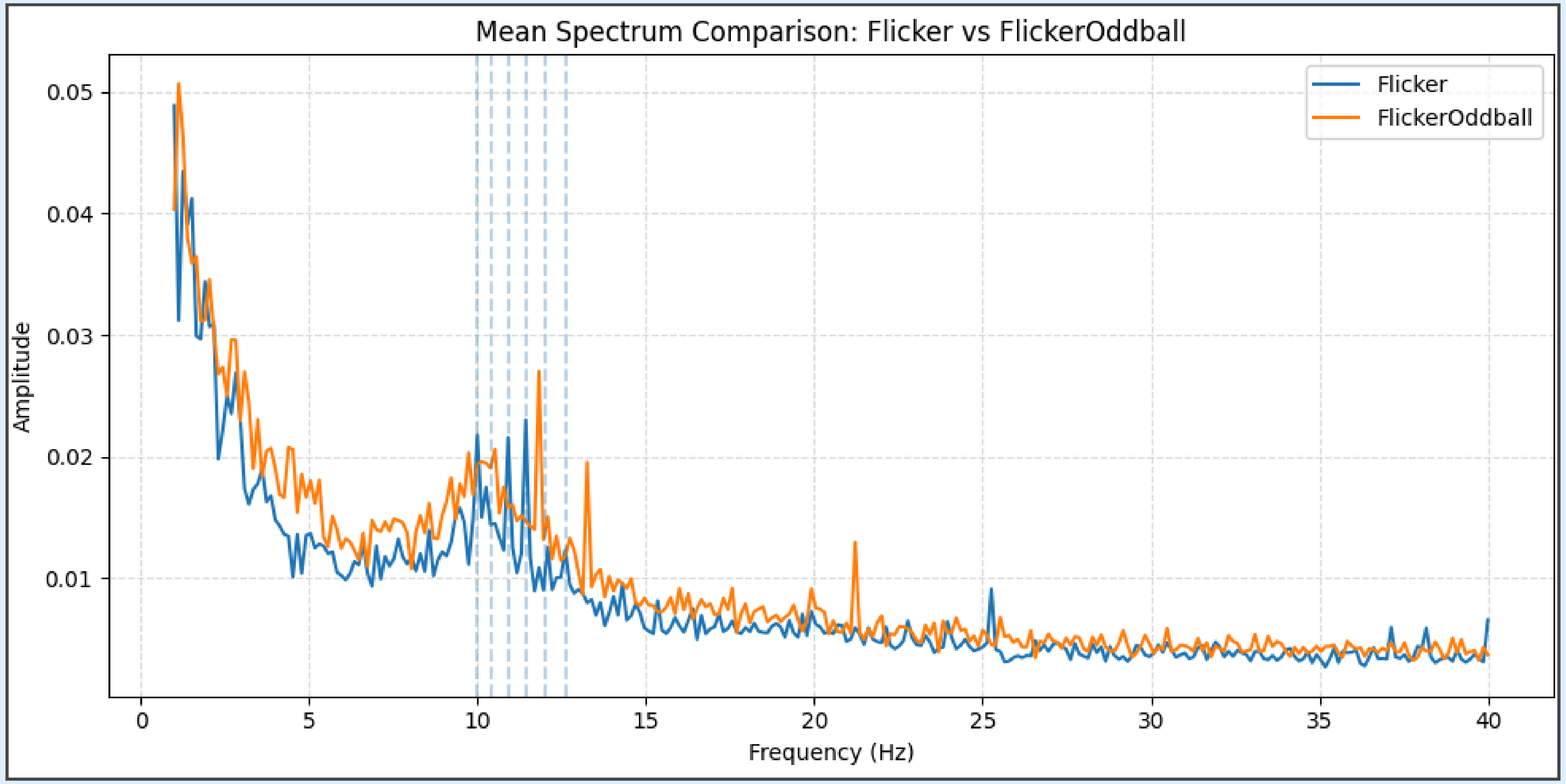


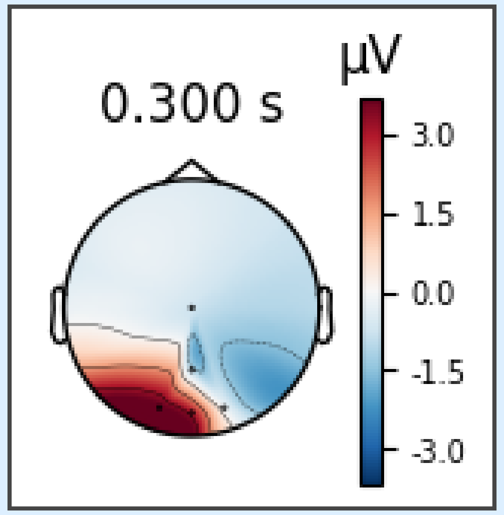
Every BCI starts with noisy data.

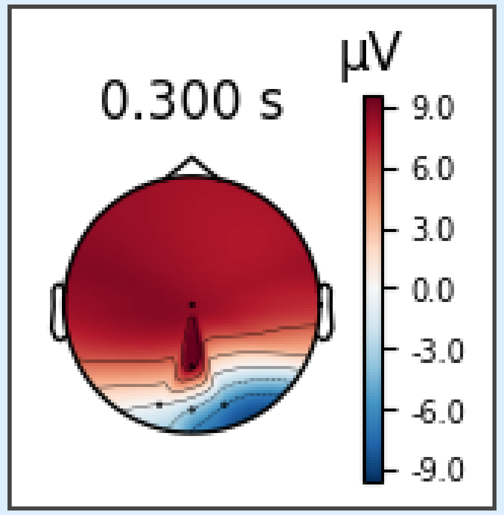
The brain is loud. Blinks. Muscle twitches. Electrode drift. 60Hz electrical hum.

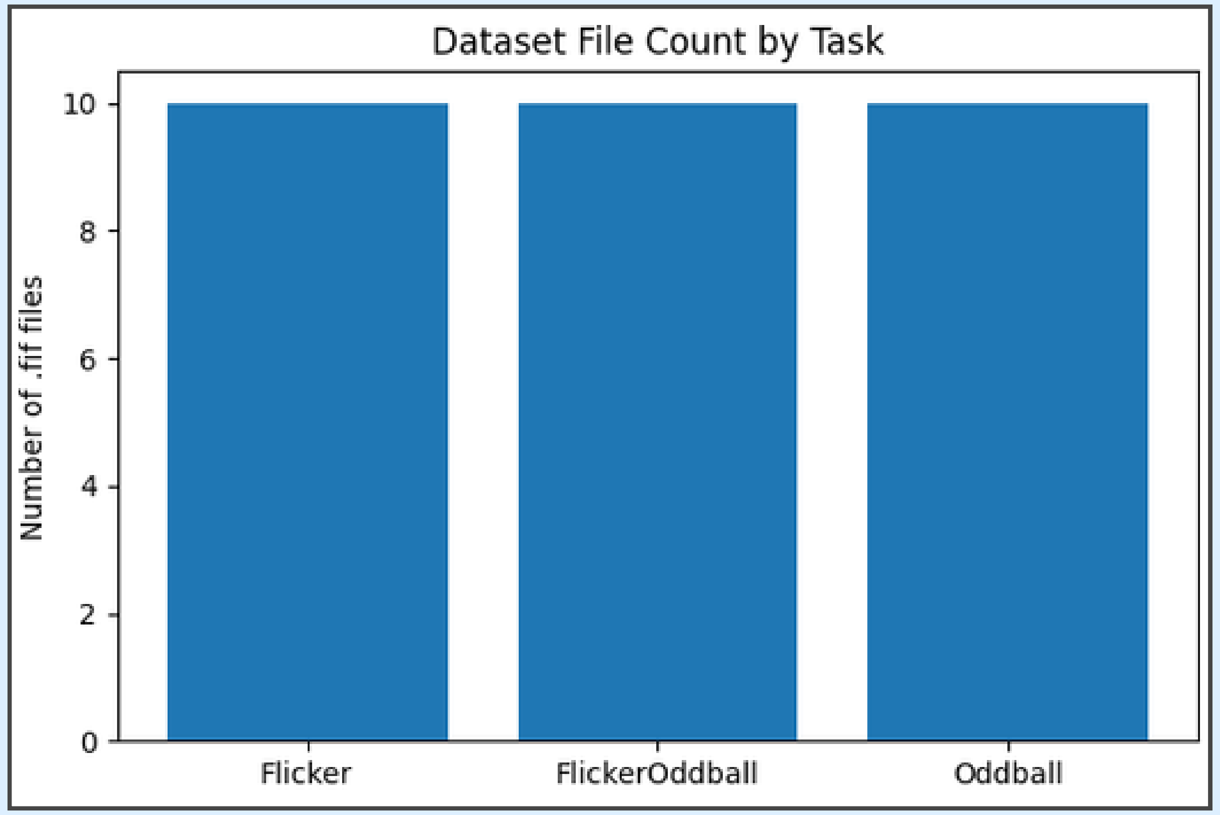
Before any classification happens, you need to separate neural signal from noise

Dataset Details









- *Some data visualizations from our dataset*

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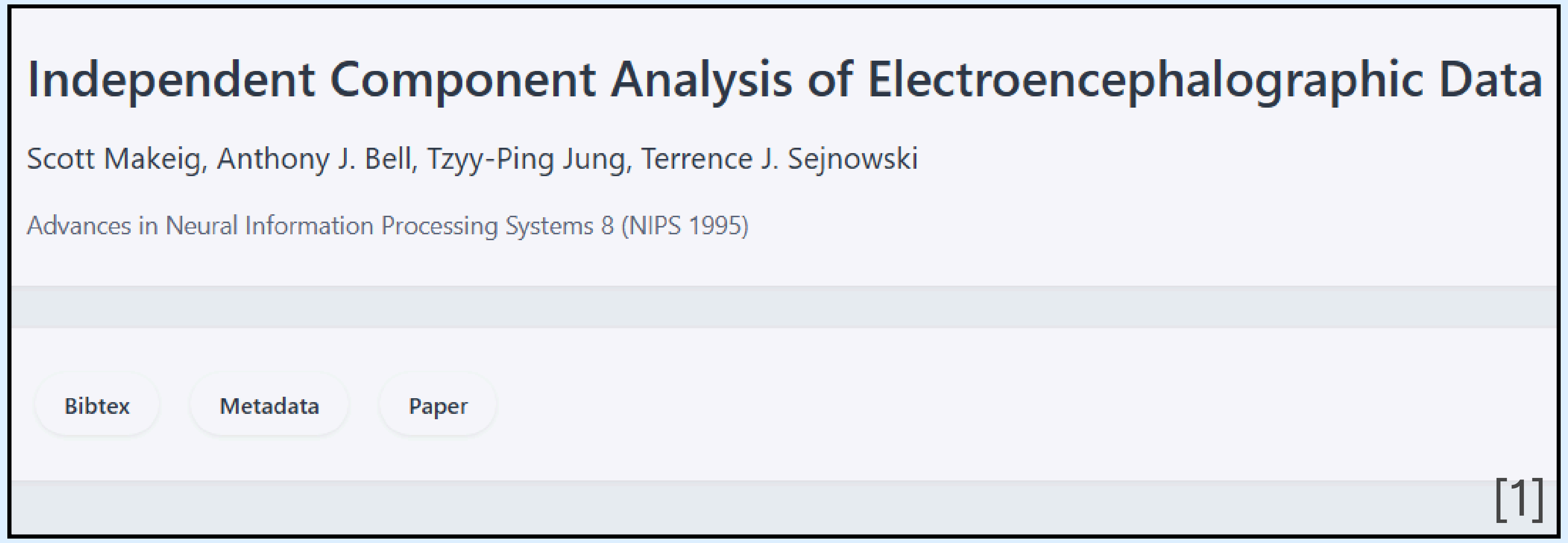
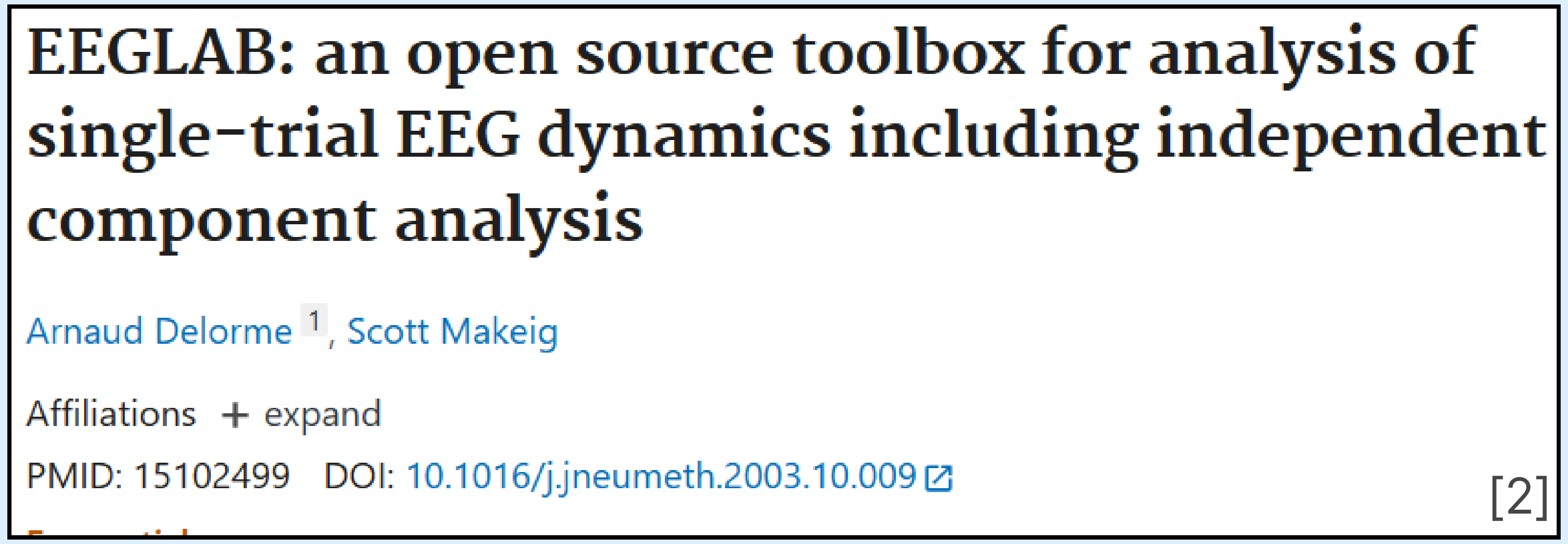

Independent Component Analysis of Electroencephalographic Data

Scott Makeig, Anthony J. Bell, Tzyy-Ping Jung, Terrence J. Sejnowski

Advances in Neural Information Processing Systems 8 (NIPS 1995)

Bibtex Metadata Paper

[1]

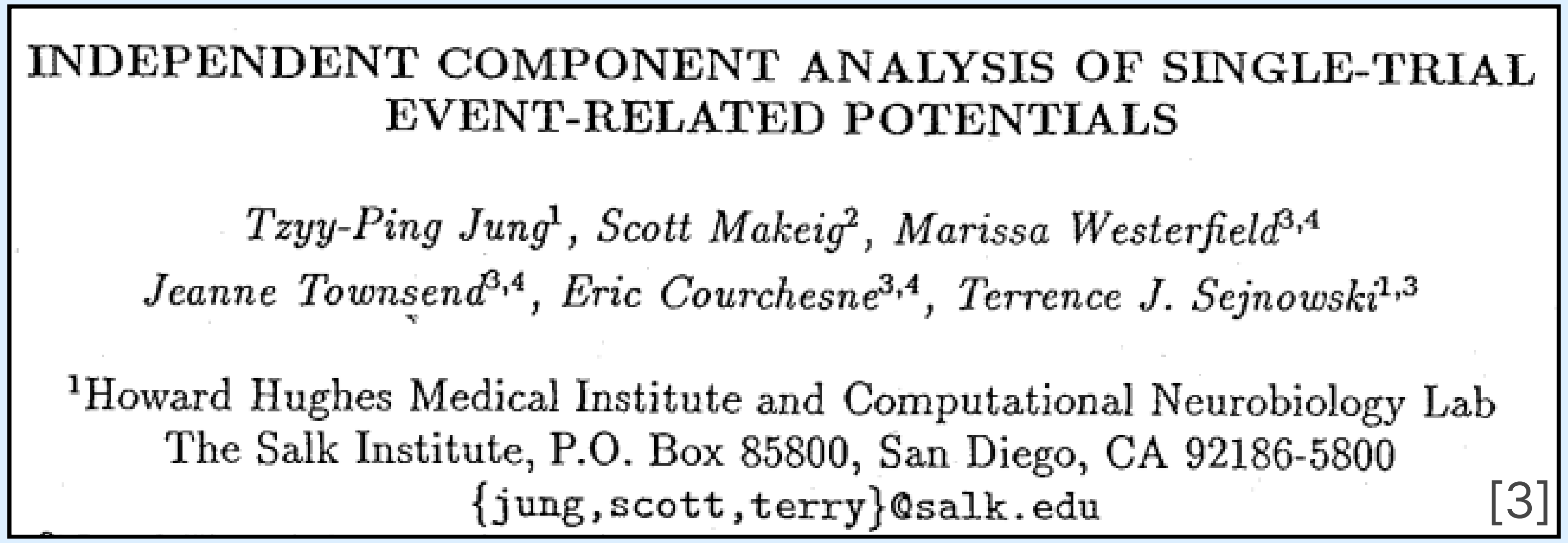

EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis

Arnaud Delorme[1], Scott Makeig

Affiliations + expand

PMID: 15102499 DOI: 10.1016/j.jneumeth.2003.10.009

[2]


INDEPENDENT COMPONENT ANALYSIS OF SINGLE-TRIAL EVENT-RELATED POTENTIALS

Tzyy-Ping Jung[1], *Scott Makeig*[2], *Marissa Westerfield*[3,4]
Jeanne Townsend[3,4], *Eric Courchesne*[3,4], *Terrence J. Sejnowski*[1,3]

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[3]

The field runs on ICA

Independent Component Analysis decomposes EEG into statistically independent sources, then removes the ones that look like artifacts.

- Used in virtually every EEG preprocessing pipeline
- Backed by 30+ years of neuroscience research
- Works remarkably well... under the right conditions

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Variability of ICA decomposition may impact EEG signals when used to remove eyeblink artifacts

Martin Billinger [2], Clemens Brunner [3]

[4]

ICA removes too much, too blindly

Gap	B1_ICA	wICA/RELAX	ASR	Bailey2025
Removes only during detected artifact events	No	Partial	No	Yes
Restricts removal to artifact-characteristic frequencies	No	Partial	No	Yes (hardcoded)
Frequencies derived from IC's own data per-subject	No	No	No	No
IC-level probability weighting (graded, not binary)	No	No	No	No
No hardcoded Hz values	No	No	N/A	No

- **The common failure:** every method either over-removes (wiping signal outside artifact windows) or relies on hardcoded frequency assumptions that don't adapt to the individual.

We address all of these gaps simultaneously.

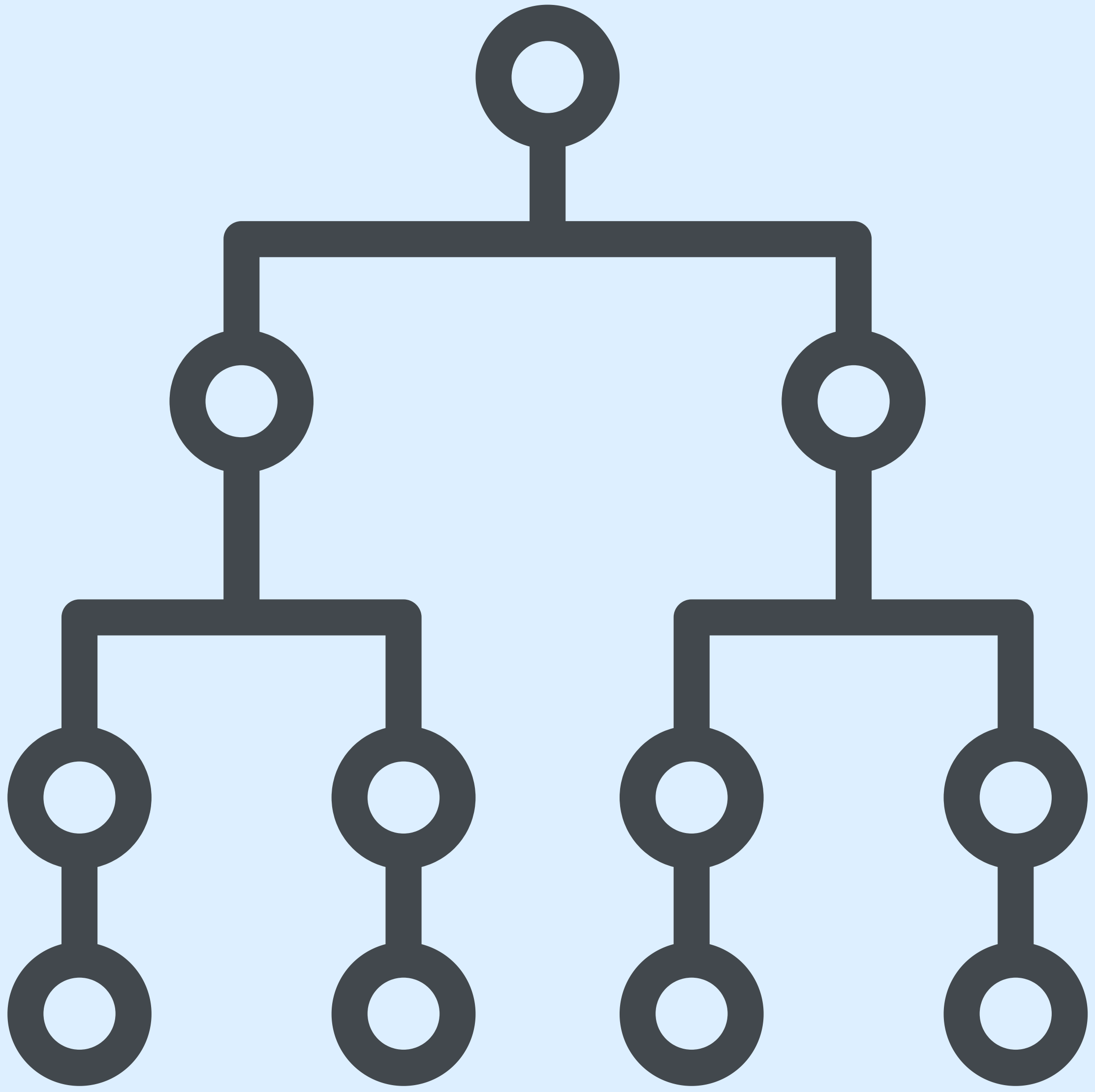
SPECTRA-ICA

Step 1: Classify

ICA decomposes the signal into 20 components. ICLabel assigns each one a 7-class probability vector.

- P(brain) + P(other) ≥ 0.50 → untouched
- Everything else → artifact pipeline
- Removal scaled by confidence: no binary reject/keep

A component that's 60% eye artifact gets treated differently than one that's 99% eye artifact.



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SPECTRA-ICA

Step 2: SPECTRA

Temporal mask: detect the actual artifact events (blink peaks, R-peaks, muscle bursts) and build a soft Gaussian window around each one. Only those windows get cleaned.

Spectral profile: compare power during artifact windows vs. outside them. Find which frequencies are elevated above the mean gain — those are the artifact-characteristic frequencies.

Blinks raise all frequencies uniformly due to the corneoretinal dipole. SPECTRA corrects for this: it only removes frequencies elevated beyond that baseline lift, protecting alpha and SSVEP bands.

3.4 Relative Spectral Excess Profiling (SPECTRA step)

This is the core novel contribution.

For eye and heart ICs (whose artifact frequency profile is physiologically well-structured), the temporal mask partitions Short-Time Fourier Transform (STFT) frames into confirmed-artifact and confirmed-clean windows. The power spectral density is estimated separately in each partition:

$$\bar{P}_{\text{art}}(k) = \frac{1}{|A|} \sum_{t \in A} |Z(k,t)|^2$$

$$\bar{P}_{\text{clean}}(k) = \frac{1}{|C|} \sum_{t \in C} |Z(k,t)|^2$$

where $A = \{t : M(t) > 0.5\}$, $C = \{t : M(t) < 0.1\}$, and $Z(k,t)$ is the STFT of the IC source.

The ratio per frequency bin is:

$$r(k) = \frac{\bar{P}_{\text{art}}(k)}{\bar{P}_{\text{clean}}(k) + \epsilon}$$

SPECTRA–ICA

Step 3: Targeted Removal

Frequency domain: filter the component's contribution down to artifact-characteristic frequencies only

Time domain: multiply by the soft temporal mask, scaled by ICLabel confidence

w(t) = p_dom · M(t)
data_clean = data_orig – Σ removal_per_IC

Fallbacks built in: if the spectral profile is too diffuse (Gini < 0.05) → broadband temporal removal. If artifact covers >60% of recording → full component subtraction.

3.5 Frequency-Selective + Temporal-Gated Removal

Given the spectral profile $F(k)$ and temporal mask $M(t)$, the remov
steps:

Step 1 (frequency domain): The IC channel-space contribution
next power of 2 and transformed via the Fast Fourier Transform (FF
interpolated to the FFT frequency grid, then inverse-FFT-transforn
to the artifact-characteristic frequencies only.

Step 2 (time domain): The frequency-filtered contribution is n
weight $w(t) = p_{\mathrm{dom}} \cdot M(t) \in [0, p_{\mathrm{dom}}]$. This gates the removal to con
the ICLabel component probability.

The decoupled formulation avoids iSTFT reconstruction artefacts t
gain adjustment (blocking/musical noise), as noted in Castellanos a
methods.

Final reconstruction: `data_clean = data_orig - sum_over_artif`

Evaluation Metrics

Pipeline	OCI	FC-OCI	SER (dB)	ARR (dB)	HF noise	alpha_ratio	seg_corr	ParetoFC
SPECTRA	**0.175**	0.107	**10.69**	-2.83	0.4801	**0.964**	**0.957**	**0.294**
Bailey2025	0.047	**0.052**	7.68	-7.84	0.5625	0.983	0.912	0.293
TAME_PT	0.211	0.125	6.27	2.34	0.4671	0.934	0.837	0.302
ASR	0.185	0.085	5.47	7.52	0.5368	0.931	0.784	0.309
TAME_ASR	0.223	0.136	6.03	5.66	0.4515	0.931	0.828	0.307
MWF	0.085	0.081	60.00	15.21	0.5581	0.965	1.000	0.329
wICA	0.080	0.096	-4.60	-5.14	0.7489	1.000	0.439	0.418
RELAX_full	0.134	0.137	-4.36	5.40	0.6630	0.964	0.451	0.408
B0_none	0.000	0.000	60.00	0.00	0.6463	1.000	1.000	0.323
B1_ICA	0.499	0.461	2.56	8.91	0.1329	0.799	0.614	0.662

- **Bold** = best value among actual cleaning methods (excluding B0_none baseline).

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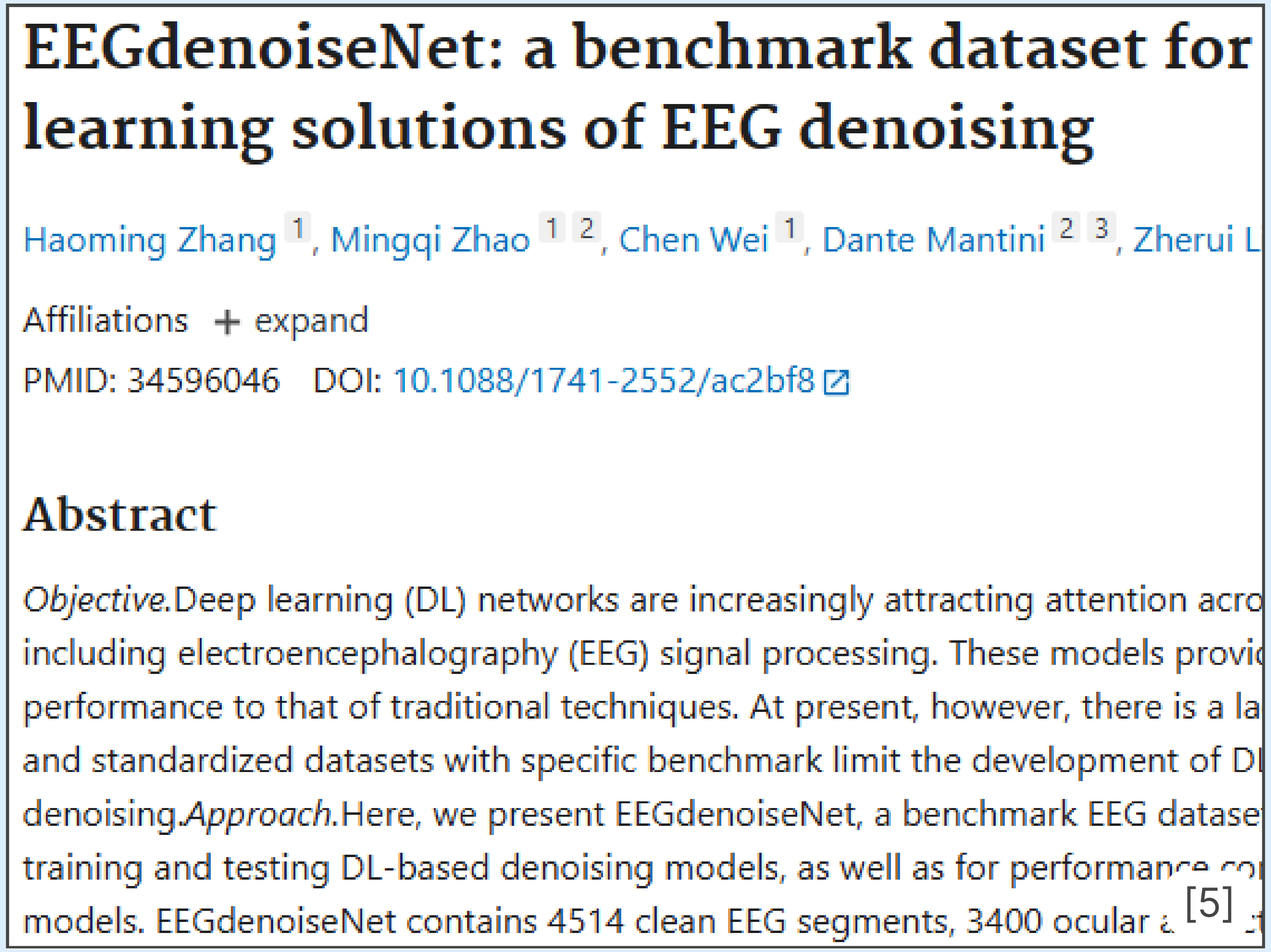
Using ICA, SPECTRA-ICA and Classification models

Does cleaner preprocessing actually help classification?

Metric	ICA	SPECTRA-ICA	Δ	Improve
Accuracy	89.54%	90.08%	+0.54pp	+0.6%
Balanced Acc	81.46%	82.72%	+1.26pp	+1.5%
Sensitivity	69.33%	71.67%	+2.33pp	+3.4%
Kappa	0.626	0.647	+0.021	+3.4%
AUC	0.900	0.911	+0.011	+1.2%

SPECTRA-ICA improves downstream classification — not just signal metrics — because it retains neural signal that ICA was discarding.

Useful beyond this dataset?



EEGdenoiseNet: a benchmark dataset for learning solutions of EEG denoising

Haoming Zhang [1], Mingqi Zhao [1,2], Chen Wei [1], Dante Mantini [2,3], Zherui L

Affiliations + expand

PMID: 34596046 DOI: 10.1088/1741-2552/ac2bf8

Abstract

*Objective.*Deep learning (DL) networks are increasingly attracting attention acro including electroencephalography (EEG) signal processing. These models provi performance to that of traditional techniques. At present, however, there is a la and standardized datasets with specific benchmark limit the development of D denoising.*Approach.*Here, we present EEGdenoiseNet, a benchmark EEG datase training and testing DL-based denoising models, as well as for performance co models. EEGdenoiseNet contains 4514 clean EEG segments, 3400 ocular a

[5]

- The core problem SPECTRA-ICA solves isn't dataset-specific.

Binary ICA destroys ~50% of neural signal power on average.

SPECTRA-ICA, from the benchmark:

- Best signal preservation in clean windows: SER 10.69 dB vs. 7.68 dB (next best)
- wICA and RELAX introduce more distortion than the uncleaned signal (negative SER)
- Alpha band preserved at 96.4%: the band most pipelines sacrifice

raw_clean, info = spectra_clean(raw, ica=ica, labels=labels, probs_matrix=probs)

Any MNE-compatible dataset. Any sampling rate. Any channel count.

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References :

- [1] Independent Component Analysis of Electroencephalographic Data (1996)
- [2] EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis (2004)
- [3] Independent Component Analysis of Single-Trial Event-Related Potentials (1999)
- [4] Variability of ICA decomposition may impact EEG signals when used to remove eyeblink artifacts (2017)
- [5] EEGdenoiseNet: a benchmark dataset for Deep Learning solutions of EEG denoising

- **Additional references are included in our report.**

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Thankyou

– Anand, Het, Nubah, Rafat, Milad